FILE TITLE **NGPDU Overheating Problem Troubleshooting** GEHW-SN112029

Restricted to General Electric Medical Systems (GEMS) personnel only.

Permission to use this Restricted Service Software and related documentation (hereinafter called the ”Material”) by persons other than GE Medical Systems employees is prohibited. This Material is the property of GEMS and contains information which is restricted to use by GEMS Manufacturing, Engineering and Service personnel. Any re-production in whole or in part is strictly prohibited. Neither this material nor any information contained herein shall be disclosed to anyone outside of GEMS Manufacturing, Engineering and Service personnel.

EFFECTIVITY:

PDU Type	Part Number
NGPDU	2326492
NGPDU-2	2326492-2
NGPDU-3	2326492-3
NGPDU-4	2326492-4

PROBLEM:

Starting from September 7th, 2005, forward production for NGPDU has started to implement overheat protection circuit to protect possible burning of the discharge resistors. The first unit that has this protection circuit for NGPDU-4 is E4YP05408, for NGPDU-3 is E4RP051167. FMI is scheduled to implement this protection circuit for IB too.

If the QB1 on the NGPDU control board (2334820-2) have short circuit, the discharge resistors will overheat and the protection thermal switch will be opened and cut off the HV DC, then following phenomenon may occurred during scan:
System cannot go to next step after clicking **Confirm** during scan, but return to **Resume** state. Message is prompted: “Scanner Hardware stopped scan.” **Start Scan** on the keyboard keeps lighting.

PURPOSE:

The purpose of this Service Note is to give the Field Engineer recommended procedure of solving above problems.

TOOLS REQUIRED:

None

SOLUTION:

DANGER

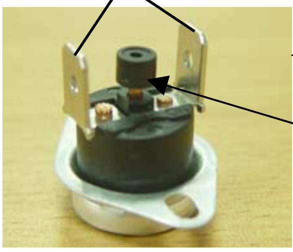
Electrocution hazard

High Voltages present.

Use proper lockout/tagout procedures before Servicing PDU.

1. Shut down the system, power off NGPDU. Cut off the power from power distribution box and use proper Lockout/Tagout procedures.
2. Use the DVM to test the two pins of Thermal Switch.
   If they're open, it indicates that the QB1 on the control board has been short-circuited.

Pins (test points)

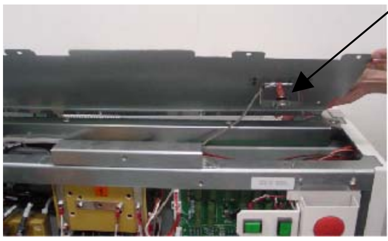


Thermal Switch

Reset Switch

3. Replace Control Board of NGPDU.
4. Press down the reset switch on the thermal switch. Use the DVM to test the two pins of the thermal switch to make sure it's closed.

Thermal switch location



Press down the reset switch directly through the hole